

# Airport Network

Design and simulation in Cisco Packet Tracer

Computer Networks | Assignment 10 Case Study

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# Objective and design approach

The airport's departments must talk to each other while staying separated and secure. The network also has to carry staff and passenger Wi-Fi, host shared services (DNS, Web, FTP, Email) and manage IP addresses centrally.

## 1 Segmentation

9 VLANs keep Admin, Check-in, Security, Baggage, Operations, IT, Wi-Fi and Servers apart

## 2 Inter-VLAN routing

R1-Core routes between VLANs using 802.1Q sub-interfaces (router-on-a-stick)

## 3 Addressing

One /24 per VLAN, DHCP pools on R1, static IPs for the servers

## 4 Shared services

DNS, Web, FTP and Email hosted together in server VLAN 90

## 5 Wireless

Two access points: Airport-Staff (VLAN 70) and Airport-Passenger (VLAN 80)

## 6 Routing

OSPF between R1-Core and R2-Edge over a 10.0.0.0/30 link

# VLAN and IP addressing plan

| VLAN | Department           | Subnet          | Gateway      |
|------|----------------------|-----------------|--------------|
| 10   | Administration       | 192.168.10.0/24 | 192.168.10.1 |
| 20   | Check-in & Ticketing | 192.168.20.0/24 | 192.168.20.1 |
| 30   | Security             | 192.168.30.0/24 | 192.168.30.1 |
| 40   | Baggage Handling     | 192.168.40.0/24 | 192.168.40.1 |
| 50   | Airport Operations   | 192.168.50.0/24 | 192.168.50.1 |
| 60   | IT Department        | 192.168.60.0/24 | 192.168.60.1 |
| 70   | Staff Wi-Fi          | 192.168.70.0/24 | 192.168.70.1 |
| 80   | Passenger Wi-Fi      | 192.168.80.0/24 | 192.168.80.1 |
| 90   | Servers              | 192.168.90.0/24 | 192.168.90.1 |

## Servers (VLAN 90)

192.168.90.10   DNS  
192.168.90.20   Web portal  
192.168.90.30   FTP + Email

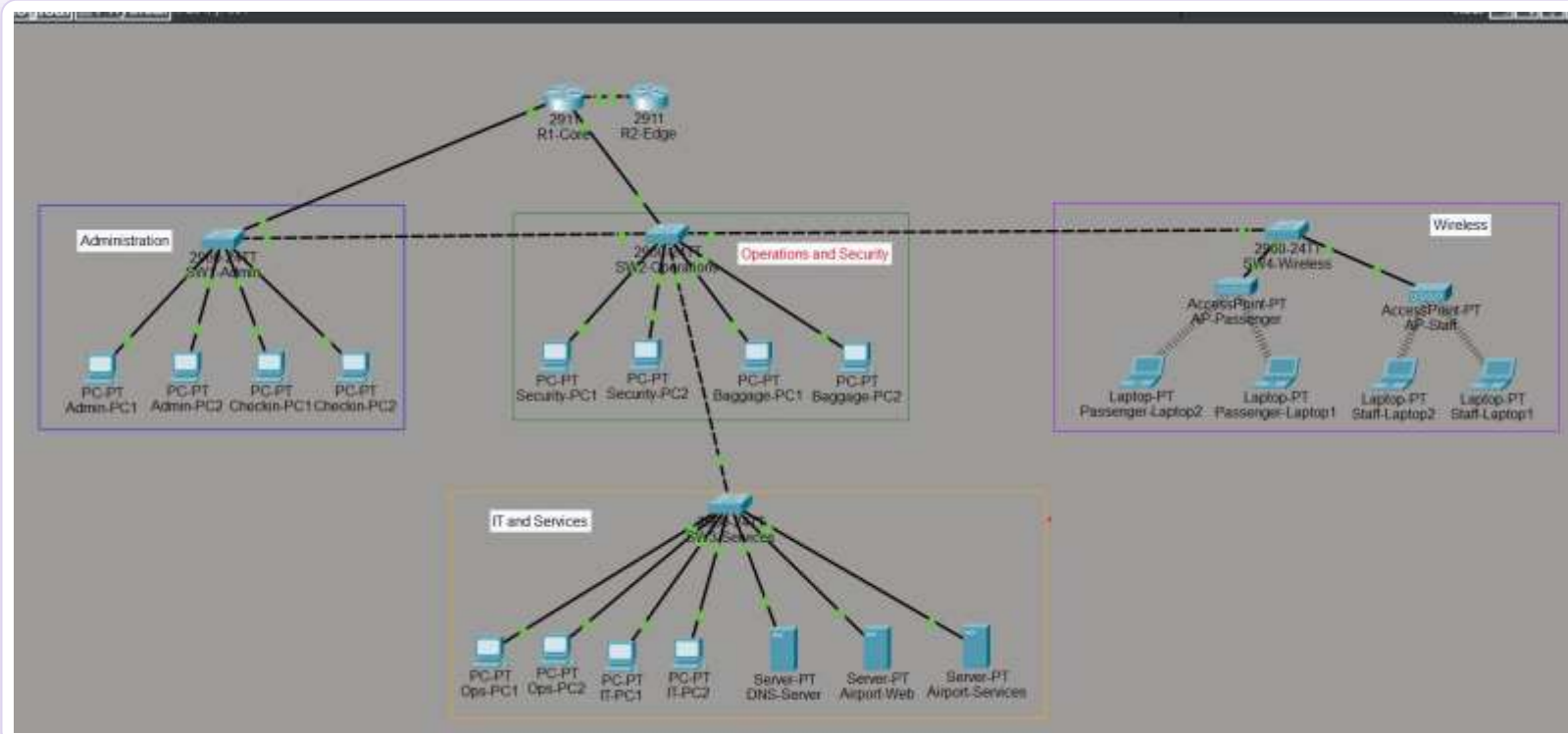
## Router link

10.0.0.0/30   R1 .1 | R2 .2

## Why this plan

One /24 per VLAN (254 hosts) keeps the plan simple. .1 is the gateway; .1 to .10 are kept out of the DHCP pools.

# Network topology



## R1-Core

Inter-VLAN routing, DHCP and OSPF

## R2-Edge

OSPF neighbour; edge router for Internet access

## SW1 to SW4

One switch per area: Admin, Operations, IT & Services, Wireless

## AP-Staff / AP-Passenger

Separate SSIDs mapped to VLAN 70 and VLAN 80

# Core routers: R1-Core and R2-Edge

## R1-Core: Gi0/0 = 10.0.0.1/30

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
```

## R2-Edge: Gi0/0 = 10.0.0.2/30 (up/up)

```
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief

Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.0.0.2        YES manual up          up
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down

Router#show ip interface brief

Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.0.0.2        YES manual up          up
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down

Router#ping 10.0.0.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/7 ms

Router#
```

### Point-to-point link

10.0.0.0/30 between R1 Gi0/0 and R2 Gi0/0

### Link proven by OSPF

R2 learns every VLAN subnet from R1 (see routing slide)

### Spare ports on R2

Gi0/1 and Gi0/2 are free for the ISP uplink

# VLANs and access ports on SW1 to SW4

## SW1-Admin: VLAN 10 on Fa0/3-4

```
SW1-Admin#enable
SW1-Admin#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1-Admin(config)#interface range fastEthernet 0/3 - 4
SW1-Admin(config-if-range)#switchport mode access
SW1-Admin(config-if-range)#switchport access vlan 10
SW1-Admin(config-if-range)#exit
SW1-Admin(config)#do show vlan brief
```

| VLAN Name               | Status | Ports                                                                                                                                                                                                             |
|-------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 default               | active | Fa0/1, Fa0/2, Fa0/5, Fa0/6<br>Fa0/7, Fa0/8, Fa0/9, Fa0/10<br>Fa0/11, Fa0/12, Fa0/13, Fa0/14<br>Fa0/15, Fa0/16, Fa0/17, Fa0/18<br>Fa0/19, Fa0/20, Fa0/21, Fa0/22<br>Fa0/23, Fa0/24, Gig0/1, Gig0/2<br>Fa0/3, Fa0/4 |
| 10 ADMIN                | active |                                                                                                                                                                                                                   |
| 1002 fddi-default       | active |                                                                                                                                                                                                                   |
| 1003 token-ring-default | active |                                                                                                                                                                                                                   |
| 1004 fddinet-default    | active |                                                                                                                                                                                                                   |
| 1005 trnet-default      | active |                                                                                                                                                                                                                   |

```
SW1-Admin(config)#
```

## SW2-Operations: VLAN 30 (Fa0/5-6), 40 (Fa0/7-8)

```
SW2-Operations#enable
SW2-Operations(config)#interface range fastEthernet 0/7 - 8
SW2-Operations(config-if-range)#switchport mode access
SW2-Operations(config-if-range)#switchport access vlan 40
SW2-Operations(config-if-range)#exit
SW2-Operations(config)#do show vlan brief
```

| VLAN Name               | Status | Ports                                                                                                                                                                        |
|-------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 default               | active | Fa0/1, Fa0/3, Fa0/4, Fa0/5<br>Fa0/20, Fa0/11, Fa0/12, Fa0/13<br>Fa0/14, Fa0/15, Fa0/16, Fa0/17<br>Fa0/18, Fa0/19, Fa0/20, Fa0/21<br>Fa0/22, Fa0/23, Fa0/24, Gig0/1<br>Gig0/2 |
| 30 SECURITY             | active | Fa0/5, Fa0/6                                                                                                                                                                 |
| 40 BAGGAGE              | active | Fa0/7, Fa0/8                                                                                                                                                                 |
| 50 OPERATIONS           | active |                                                                                                                                                                              |
| 60 IT                   | active |                                                                                                                                                                              |
| 70 STAFF-WIFI           | active |                                                                                                                                                                              |
| 80 GUEST-WIFI           | active |                                                                                                                                                                              |
| 90 SERVERS              | active |                                                                                                                                                                              |
| 1002 fddi-default       | active |                                                                                                                                                                              |
| 1003 token-ring-default | active |                                                                                                                                                                              |
| 1004 fddinet-default    | active |                                                                                                                                                                              |
| 1005 trnet-default      | active |                                                                                                                                                                              |

```
SW2-Operations(config)#
```

## SW3-Services: VLAN 50, 60 and 90

```
SW3-Services(config)#do show vlan brief
```

| VLAN Name               | Status | Ports                                                                                                                                                 |
|-------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 default               | active | Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Fa0/24<br>Gig0/1, Gig0/2 |
| 50 OPERATIONS           | active | Fa0/2, Fa0/3                                                                                                                                          |
| 60 IT                   | active | Fa0/4, Fa0/5                                                                                                                                          |
| 90 SERVERS              | active | Fa0/6, Fa0/7, Fa0/8                                                                                                                                   |
| 1002 fddi-default       | active |                                                                                                                                                       |
| 1003 token-ring-default | active |                                                                                                                                                       |
| 1004 fddinet-default    | active |                                                                                                                                                       |
| 1005 trnet-default      | active |                                                                                                                                                       |

```
SW3-Services(config)#
```

## SW4-Wireless: VLAN 70 and 80

```
SW4-Wireless(config-vlan)#exit
SW4-Wireless(config)#do show vlan brief
```

| VLAN Name               | Status | Ports                                                                                                                                                                                                      |
|-------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 default               | active | Fa0/2, Fa0/3, Fa0/4, Fa0/5<br>Fa0/6, Fa0/7, Fa0/8, Fa0/9<br>Fa0/10, Fa0/11, Fa0/12, Fa0/13<br>Fa0/14, Fa0/15, Fa0/16, Fa0/17<br>Fa0/18, Fa0/19, Fa0/20, Fa0/21<br>Fa0/22, Fa0/23, Fa0/24, Gig0/1<br>Gig0/2 |
| 70 STAFF-WIFI           | active |                                                                                                                                                                                                            |
| 80 GUEST-WIFI           | active |                                                                                                                                                                                                            |
| 1002 fddi-default       | active |                                                                                                                                                                                                            |
| 1003 token-ring-default | active |                                                                                                                                                                                                            |
| 1004 fddinet-default    | active |                                                                                                                                                                                                            |
| 1005 trnet-default      | active |                                                                                                                                                                                                            |

```
SW4-Wireless(config)#
```

# Trunking and inter-VLAN routing

## SW2 trunks (802.1Q, native VLAN 1)

```
SW2-Operations(config-if-range)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

SW2-Operations(config)#do show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1
Fa0/2     on        802.1q         trunking      1
Fa0/3     on        802.1q         trunking      1
Fa0/4     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Fa0/4     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,30,40,50,60,70,80,90
Fa0/2     1,30,40,50,60,70,80,90
Fa0/3     1,30,40,50,60,70,80,90
Fa0/4     1,30,40,50,60,70,80,90

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,30,40,50,60,70,80,90
Fa0/2     none
Fa0/3     none
--More--
```

## R1 sub-interfaces = VLAN gateways

```
Router(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0       10.0.0.1        YES manual  up      up
GigabitEthernet0/1       unassigned      YES unset   up      up
GigabitEthernet0/1.10    192.168.10.1    YES manual  up      up
GigabitEthernet0/1.20    192.168.20.1    YES manual  up      up
GigabitEthernet0/2       unassigned      YES unset   up      up
GigabitEthernet0/2.30    192.168.30.1    YES manual  up      up
GigabitEthernet0/2.40    192.168.40.1    YES manual  up      up
GigabitEthernet0/2.50    192.168.50.1    YES manual  up      up
GigabitEthernet0/2.60    192.168.60.1    YES manual  up      up
GigabitEthernet0/2.70    192.168.70.1    YES manual  up      up
GigabitEthernet0/2.80    192.168.80.1    YES manual  up      up
GigabitEthernet0/2.90    192.168.90.1    YES manual  up      up
Vlan1                    unassigned      YES unset   administratively down down
Router(config)#
```

- R1 Gi0/1 to SW1 carries VLAN 10 and 20 (Gi0/1.10, Gi0/1.20)
- R1 Gi0/2 to SW2 carries VLAN 30 to 90 (Gi0/2.30 to Gi0/2.90)
- Each sub-interface is that VLAN's default gateway, 192.168.X.1
- Cross-VLAN pings show TTL 127: exactly one routed hop through R1

# DHCP: one pool per VLAN on R1

## Pool configuration on R1-Core

```
Router(config)#  
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10  
Router(config)#ip dhcp pool VLAN10-ADMIN  
Router(dhcp-config)#network 192.168.10.0 255.255.255.0  
Router(dhcp-config)#default-router 192.168.10.1  
Router(dhcp-config)#dns-server 192.168.90.10  
Router(dhcp-config)#exit  
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10  
Router(config)#ip dhcp pool VLAN20-CHECKIN  
Router(dhcp-config)#network 192.168.20.0 255.255.255.0  
Router(dhcp-config)#default-router 192.168.20.1  
Router(dhcp-config)#dns-server 192.168.90.10  
Router(dhcp-config)#exit  
Router(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.10  
Router(config)#ip dhcp pool VLAN30-SECURITY  
Router(dhcp-config)#network 192.168.30.0 255.255.255.0  
Router(dhcp-config)#default-router 192.168.30.1  
Router(dhcp-config)#dns-server 192.168.90.10  
Router(dhcp-config)#exit  
Router(config)#ip dhcp excluded-address 192.168.40.1 192.168.40.10  
Router(config)#ip dhcp pool VLAN40-BAGGAGE  
Router(dhcp-config)#network 192.168.40.0 255.255.255.0  
Router(dhcp-config)#default-router 192.168.40.1  
Router(dhcp-config)#dns-server 192.168.90.10  
Router(dhcp-config)#exit  
Router(config)#ip dhcp excluded-address 192.168.50.1 192.168.50.10  
Router(config)#ip dhcp pool VLAN50-OPERATIONS  
Router(dhcp-config)#network 192.168.50.0 255.255.255.0  
Router(dhcp-config)#default-router 192.168.50.1  
Router(dhcp-config)#dns-server 192.168.90.10  
Router(dhcp-config)#exit
```

## Leases received by clients

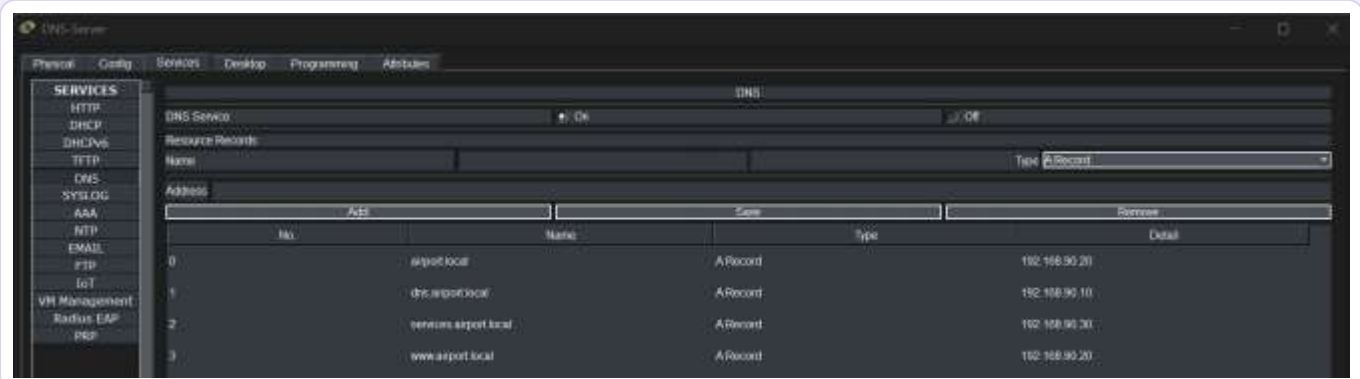
| Client        | VLAN | Leased address |
|---------------|------|----------------|
| Admin-PC1     | 10   | 192.168.10.11  |
| Checkin-PC1   | 20   | 192.168.20.11  |
| Baggage-PC1   | 40   | 192.168.40.11  |
| Ops-PC1       | 50   | 192.168.50.11  |
| IT-PC1        | 60   | 192.168.60.11  |
| Staff-Laptop1 | 70   | 192.168.70.11  |

- Each pool sets the VLAN gateway (X.1) and DNS server 192.168.90.10
- 192.168.X.1 to .10 are excluded, so the first lease is .11
- Servers use static addresses outside DHCP

# Centralised services on VLAN 90

| Server           | IP            | Role                  |
|------------------|---------------|-----------------------|
| DNS-Server       | 192.168.90.10 | DNS for airport.local |
| Airport-Web      | 192.168.90.20 | HTTP portal           |
| Airport-Services | 192.168.90.30 | FTP, SMTP and POP3    |

## DNS service: A records

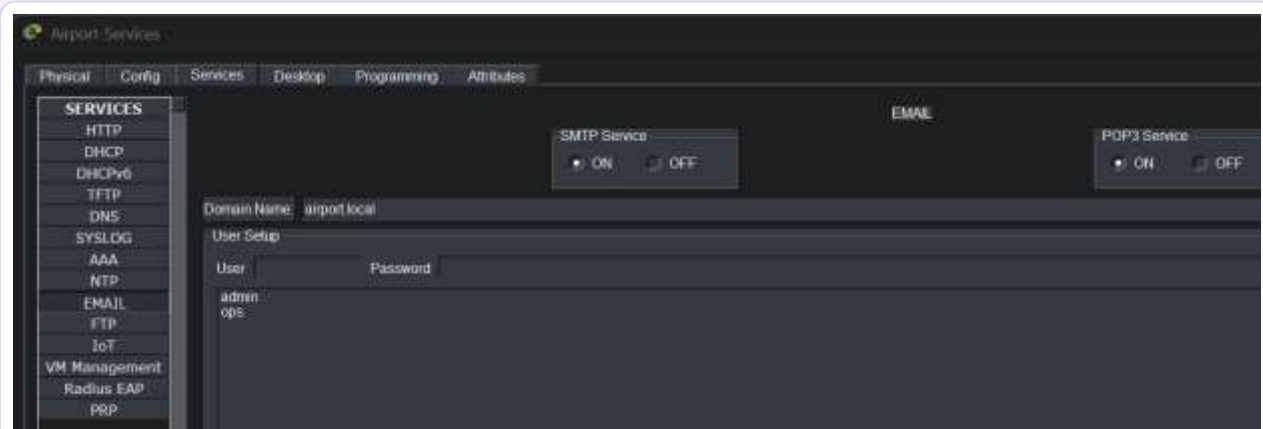


## DNS records

airport.local -> .20  
www.airport.local -> .20  
dns.airport.local -> .10  
services.airport.local -> .30

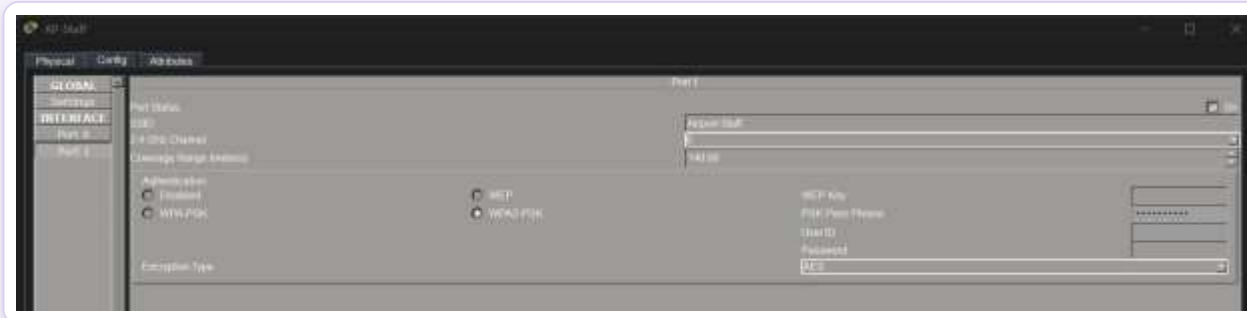
Mail domain airport.local, users admin and ops.

## Email service: SMTP + POP3 on



## Wireless: staff and passenger access

## Access point settings (passphrases masked)



## Staff-Laptop1 joins Airport-Staff



- SSIDs Airport-Staff (VLAN 70) and Airport-Passenger (VLAN 80)
- WPA2-PSK with AES, 2.4 GHz
- Staff-Laptop1 gets 192.168.70.11/24 by DHCP, gateway 192.168.70.1, DNS 192.168.90.10

Staff-Laptop1 pings its gateway 192.168.70.1: 4/4 replies, TTL 255.

# Dynamic routing with OSPF

## show ip route on R2-Edge

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
O       192.168.10.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O       192.168.20.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O       192.168.30.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O       192.168.40.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O       192.168.50.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O       192.168.60.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O       192.168.70.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O       192.168.80.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O       192.168.90.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
```

9

VLAN subnets learned from R1 through OSPF

110 / 2

Administrative distance and metric of each route

10.0.0.1

Next hop for all learned routes (R1-Core)

# Connectivity test results

| Source                      | Gateway | DNS .90.10 | Web .90.20 | Services .90.30 | Application test |
|-----------------------------|---------|------------|------------|-----------------|------------------|
| Admin-PC1 (VLAN 10)         | ✓       | ✓          | -          | -               |                  |
| Checkin-PC1 (VLAN 20)       | ✓       | ✓          | -          | -               |                  |
| Security-PC1 (VLAN 30)      | ✓       | ✓          | ✓          | -               | HTTP by name     |
| Baggage-PC1 (VLAN 40)       | ✓       | ✓          | -          | -               |                  |
| Ops-PC1 (VLAN 50)           | ✓       | ✓          | -          | -               |                  |
| IT-PC1 (VLAN 60)            | ✓       | ✓          | -          | -               |                  |
| Staff-Laptop1 (VLAN 70)     | ✓       | ✓          | ✓          | ✓               | DNS lookup, HTTP |
| Passenger-Laptop1 (VLAN 80) | ✓       | ✓          | -          | -               | HTTP by name     |
| Passenger-Laptop2 (VLAN 80) | ✓       | ✓          | -          | -               |                  |
| Airport-Services (VLAN 90)  | ✓       | -          | -          | -               |                  |
| DNS-Server (VLAN 90)        | ✓       | -          | -          | -               |                  |

✓ = 4/4 replies, 0% loss. - = not captured. TTL 255 = reply from the R1 sub-interface, TTL 127 = packet crossed R1 once (inter-VLAN routing works).

# Application-layer services

```
C:\>nslookup airport.local

Server: [192.168.90.10]
Address: 192.168.90.10

Non-authoritative answer:
Name:   airport.local
Address: 192.168.90.20

C:\>
```

## DNS (UDP 53)

nslookup airport.local returns 192.168.90.20 from the DNS server 192.168.90.10



## HTTP (TCP 80)

The airport portal opens by name from Staff, Security and Passenger devices

```
C:\>
C:\>ftp 192.168.90.30
Trying to connect...192.168.90.30
Connected to 192.168.90.30
220- Welcome to PT Ftp server
Username: .....
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>
```

## FTP (TCP 21)

Login to Airport-Services 192.168.90.30 succeeds: 230 Logged in

**Name resolution, web, file transfer and mail work across VLANs because R1-Core routes every request to server VLAN 90.**

# Simulation trace: DNS query

Simulation Panel

Event List

| Vis. | Time(sec) | Last Device    |
|------|-----------|----------------|
|      | 0.009     | —              |
|      | 0.010     | Staff-Laptop1  |
|      | 0.011     | AP-Staff       |
|      | 0.012     | SW4-Wireless   |
|      | 0.013     | SW2-Operations |
|      | 0.013     | —              |
|      | 0.014     | AP-Staff       |
|      | 0.014     | AP-Staff       |
|      | 0.014     | R1-Core        |
|      | 0.015     | SW2-Operations |
|      | 0.015     | SW2-Operations |
|      | 0.015     | SW2-Operations |
|      | 0.016     | SW3-Services   |
|      | 0.016     | SW3-Services   |
|      | 0.016     | SW3-Services   |
|      | 0.017     | DNS-Server     |
|      | 0.018     | SW3-Services   |
|      | 0.019     | SW2-Operations |

Reset Simulation ☒ Constant Delay Captured to 0.019

Play Controls

PDU Information at Device: DNS-Server

| OSI Model                                                                    | Inbound PDU Details | Outbound PDU Details                                               |
|------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------|
| At Device: DNS-Server<br>Source: Staff-Laptop1<br>Destination: 192.168.90.10 |                     |                                                                    |
| <b>In Layers</b>                                                             |                     | <b>Out Layers</b>                                                  |
| Layer 7: DNS                                                                 |                     | Layer 7: DNS                                                       |
| Layer6                                                                       |                     | Layer6                                                             |
| Layer5                                                                       |                     | Layer5                                                             |
| Layer 4: UDP Src Port: 1027, Dst Port: 53                                    |                     | Layer 4: UDP Src Port: 53, Dst Port: 1027                          |
| Layer 3: IP Header Src. IP: 192.168.70.11, Dest. IP: 192.168.90.10           |                     | Layer 3: IP Header Src. IP: 192.168.90.10, Dest. IP: 192.168.70.11 |
| Layer 2: Ethernet II Header 00E0.8FDA.C803 >> 00D0.BA19.701C                 |                     | Layer 2: Ethernet II Header 00D0.BA19.701C >> 00E0.8FDA.C803       |
| Layer 1: Port FastEthernet0                                                  |                     | Layer 1: Port(s): FastEthernet0                                    |
| 1. FastEthernet0 receives the frame.                                         |                     |                                                                    |
| Challenge Me << Previous Layer Next Layer >>                                 |                     |                                                                    |

## Path of the request

Staff-Laptop1 > AP-Staff > SW4 > SW2 > R1-Core > SW2 > SW3 > DNS-Server

## What the PDU shows

- Application: DNS
- Transport: UDP, port 1027 to 53
- Internet: 192.168.70.11 to 192.168.90.10
- Access: source MAC is R1's, because R1 routed the packet between VLANs

# Simulation trace: HTTP over TCP

PDU Information at Device: Security-PC1

OSI Model

Outbound PDU Details

At Device: Security-PC1

Source: Security-PC1

Destination: 192.168.90.20

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer1

Out Layers

Layer7

Layer6

Layer5

Layer 4: TCP Src Port: 1026, Dst Port: 80

Layer 3: IP Header Src. IP: 192.168.30.11, Dest. IP: 192.168.90.20

Layer 2: Ethernet II Header 0002.4A71.EA27 >> 00E0.8FDA.C803

Layer 1: Port(s):

1. The device closes the TCP connection to 192.168.90.20 on port 80.

2. The device sets the connection state to FIN\_WAIT\_1.

3. The device sends a TCP FIN+ACK segment.

4. Sent segment information: the sequence number 1, the ACK number 1, and the data length 20.

Challenge Me

<< Previous Layer

Next Layer >>

## Security-PC1 to Airport-Web

- Transport: TCP, source port 1026 to destination port 80
- Internet: 192.168.30.11 to 192.168.90.20
- Access: Ethernet II frame addressed to R1, the default gateway
- Captured here: connection teardown (FIN+ACK, state FIN\_WAIT\_1)

## Life of a TCP connection

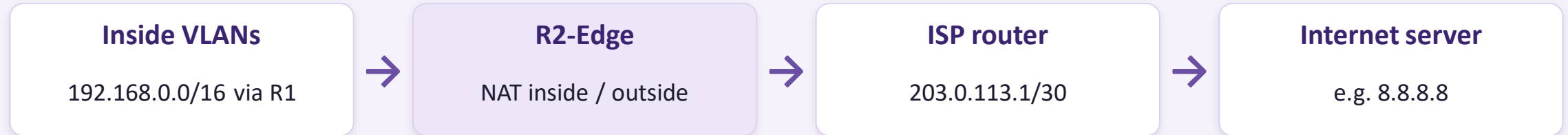


Highlighted step = the segment in the screenshot.

# TCP/IP layers in our network

| Layer          | Protocols used                                                   | Evidence from our tests                                                 |
|----------------|------------------------------------------------------------------|-------------------------------------------------------------------------|
| Application    | DNS, HTTP, FTP, SMTP/POP3, DHCP                                  | nslookup airport.local = 192.168.90.20; portal page loads               |
| Transport      | UDP: DNS 53, DHCP 67/68. TCP: HTTP 80, FTP 21, SMTP 25, POP3 110 | DNS: UDP 1027 to 53. HTTP: TCP 1026 to 80                               |
| Internet       | IPv4, ICMP, ARP, OSPF                                            | 192.168.70.11 to 192.168.90.10; pings with TTL 127; OSPF routes [110/2] |
| Network Access | Ethernet II, 802.1Q trunks, 802.11 (WPA2-PSK)                    | Source MAC rewritten by R1; SW2 trunks; Airport-Staff SSID              |

# Internet access: NAT/PAT on R2-Edge



```
! R2-Edge
interface GigabitEthernet0/0
 ip nat inside
interface GigabitEthernet0/1
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 no shutdown
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface Gi0/1 overload
ip route 0.0.0.0 0.0.0.0 203.0.113.1
router ospf 1
 default-information originate
```

*Add screenshot: show ip nat translations on R2-Edge  
after a staff PC pings the Internet server*

- PAT shares one public address for all VLANs
- R1 learns the default route through OSPF

# Security: isolating guest Wi-Fi with an ACL

Today: guests can reach the DNS server

```
C:\>ping 192.168.90.10

Pinging 192.168.90.10 with 32 bytes of data:

Reply from 192.168.90.10: bytes=32 time=25ms TTL=127
Reply from 192.168.90.10: bytes=32 time=23ms TTL=127
Reply from 192.168.90.10: bytes=32 time=15ms TTL=127
Reply from 192.168.90.10: bytes=32 time=24ms TTL=127

Ping statistics for 192.168.90.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 25ms, Average = 21ms

C:\>
```

```
ip access-list extended GUEST-IN
 permit udp any host 192.168.90.10 eq 53
 permit tcp any host 192.168.90.20 eq 80
 deny    ip any 192.168.0.0 0.0.255.255
 permit ip any any
!
interface GigabitEthernet0/2.80
 ip access-group GUEST-IN in
```

*Add screenshot: Passenger-Laptop1 ping to 192.168.90.30 fails,  
while http://airport.local still opens*

- Guests keep DNS, the web portal and the Internet
- All other internal subnets are blocked

# Conclusion

9

VLANs, one per department or service

2 + 4 + 2

routers, switches and access points

4

services: DNS, Web, FTP, Email

0%

packet loss in every captured test

Thank you